



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SUBHA MITRAA FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

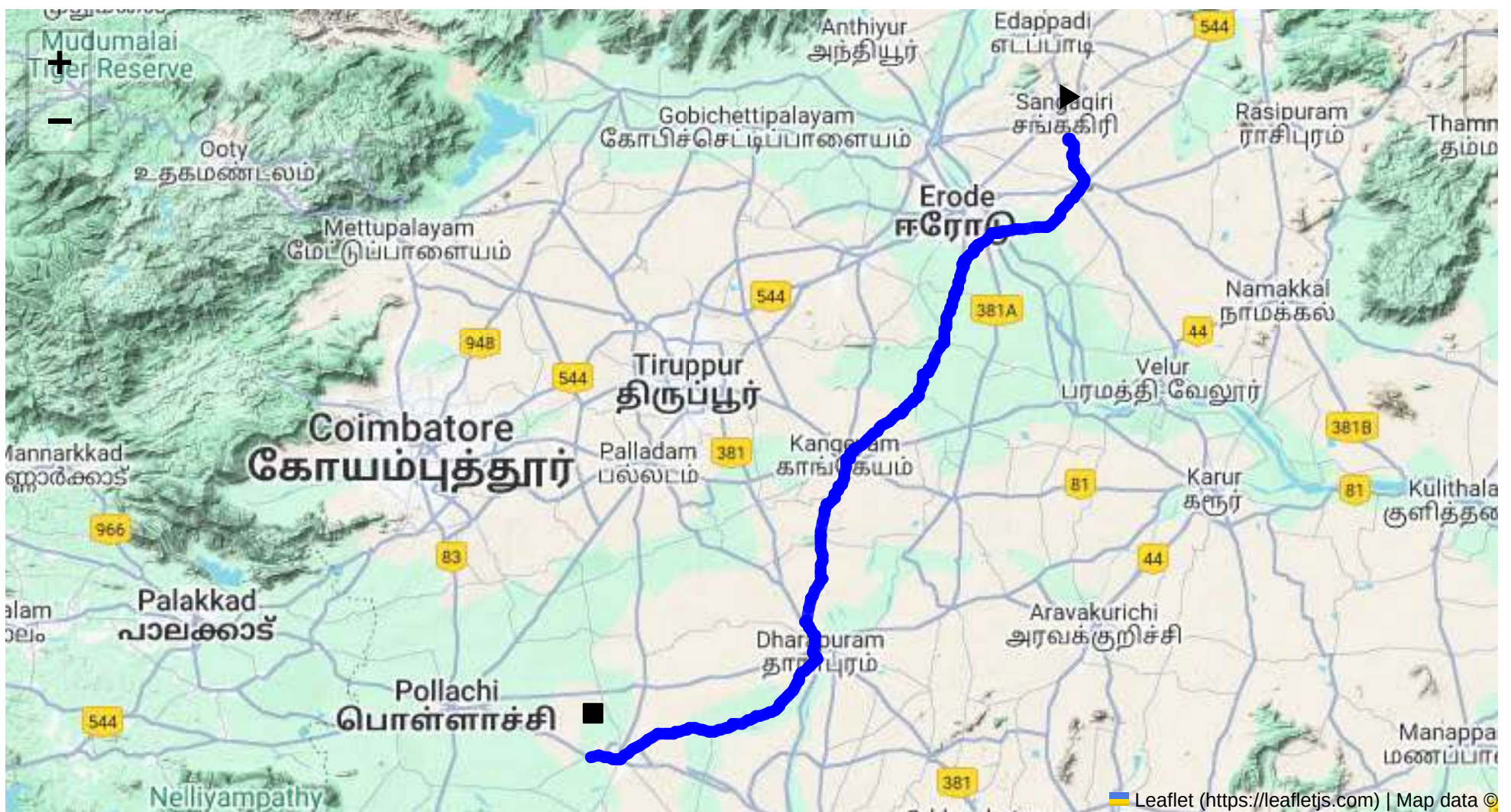
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 144.49 km**  
**Estimated Duration: 3.2 hours**  
**Adjusted Duration (Heavy Vehicle): 4.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.591277, 77.207579)**



## Welcome to the Journey Risk Management Study

1. **Overview of the Route Map:** The route spans approximately 144.49 kilometers and mainly travels through rural and semi-urban areas. It begins at Sangagiri, traveling through Tiruchengode, continuing via Udumalpet - Dharapuram Road, and concludes near NH209 in Tamil Nadu. The path is characterized by a mix of state and national highways, and local roads. Given the transition through agricultural and small-town areas, diversions and occasional localized routes are expected.



2. **Typical Weather Conditions and Potential Weather-Related Hazards:** The region experiences a tropical climate with the potential for hot and dry conditions during summers (March to May), and significant rainfall during the monsoon season (June to September). This can lead to slippery roads, reduced visibility, and flooding in low-lying areas, particularly between Udumalpet and Dharapuram.
3. **Analysis of Traffic Patterns:** Traffic peaks usually coincide with school and work commute times: early morning (7-9 AM) and late afternoon (4-6 PM). Roads near Tiruchengode and Udumalpet might experience congestion during these peak times, with additional delays due to local markets and festivals. Traffic density is lower in rural stretches but can suddenly increase near towns and urban areas.
4. **Assessment of Road Quality and Infrastructure:** The majority of the route is on well-paved roads. However, periodic maintenance might result in unpaved sections or detours. Some areas may have narrow roads, especially in village sections, which could pose challenges for heavy vehicles. Road signage and lighting are generally adequate, but caution should be exercised on less-marked local roads.
5. **Suggestions for Alternative Routes for Emergencies:** In case of blockages, alternate routes include local roads parallel to the main highways. For major detours, using NH544 via Erode to NH209 is advisable, although this might increase travel time. Local knowledge is crucial for navigating smaller bypasses efficiently.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Transport of hazardous materials may be restricted during certain hours in populated areas, typically avoiding peak congestion periods. Compliance with licensing and signage regulations is mandatory, and carrying appropriate documentation is essential. Restrictions on entering certain town centers should also be reviewed in advance.
7. **Overview of Historical Incidents:** Historical data indicates occasional road accidents involving heavy vehicles; however, these mostly occurred due to driver fatigue and poor road conditions during monsoons. Hazardous material incidents are rare but tend to be concentrated around more populated areas where traffic is dense.
8. **Environmental Considerations and Sensitive Areas:** The route travels through agricultural areas and vicinity forests near the Western Ghats, which are environmentally sensitive. Efforts should be made to minimize pollution and potential contamination of soil and water bodies near these regions. Noise pollution should also be considered when traveling through quieter towns.
9. **Analysis of Communication Coverage:** Most of the route is well-covered by mobile networks, although there are potential communication dead zones around forested or mountainous parts near the Western Ghats. Ensuring continuous communication availability via multi-network SIM cards is advisable.
10. **Estimated Emergency Response Times:** In town areas, emergency services might reach within 30 minutes. In more rural or mountainous areas, response times can extend to 1-2 hours depending on proximity to significant urban centers.
11. **Overall Summary of Risk Assessment:** Overall, the route presents moderate risk due to weather variability and road conditions, balanced by generally good road infrastructure. Key risks include

weather disruptions, traffic congestion in towns, and potential communication gaps. Mitigation strategies should focus on timing travel to avoid peak congestion and ensuring preparedness for weather-related issues. Transporters should remain informed of local regulations and maintain a flexible approach to route adjustments as needed.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Roundabout | High       | 11.00172, 77.56268 | 15 KM/Hr    |
| 2  | Roundabout | High       | 10.72668, 77.52217 | 15 KM/Hr    |
| 3  | Roundabout | High       | 10.58763, 77.24722 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 5  | Turn       | High       | 11.44022, 77.87546 | 15 KM/Hr    |
| 6  | Turn       | Medium     | 11.43107, 77.88171 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.39976, 77.88382 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.39690, 77.88852 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.36593, 77.88847 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.33561, 77.86176 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.31790, 77.84330 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.30951, 77.77598 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.31109, 77.77183 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.30911, 77.76654 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.29496, 77.74717 | 15 KM/Hr    |
| 16 | Turn       | High       | 11.29128, 77.74656 | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.28859, 77.74323 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.28533, 77.74247 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.27310, 77.72938 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.27330, 77.72720 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.23062, 77.71799 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.15942, 77.70068 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.12126, 77.67929 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.11935, 77.67790 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.11765, 77.67237 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 26 | Turn       | Medium     | 11.11660, 77.67164 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.10462, 77.67055 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.08701, 77.66412 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.06229, 77.63986 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.00834, 77.56962 | 30 KM/Hr    |
| 31 | Blind Spot | Blind Spot | 11.00065, 77.56884 | 10 KM/Hr    |
| 32 | Turn       | Medium     | 10.94873, 77.54679 | 30 KM/Hr    |
| 33 | Turn       | High       | 10.94732, 77.54710 | 15 KM/Hr    |
| 34 | Turn       | Medium     | 10.93774, 77.53723 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 10.92953, 77.53274 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 10.89225, 77.52651 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 10.83362, 77.52899 | 30 KM/Hr    |
| 38 | Turn       | High       | 10.77658, 77.50715 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 10.75748, 77.51939 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 10.73682, 77.51732 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 10.63697, 77.41792 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 10.63715, 77.40497 | 30 KM/Hr    |
| 43 | Turn       | Medium     | 10.62891, 77.38715 | 30 KM/Hr    |
| 44 | Turn       | Medium     | 10.62505, 77.37856 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 10.63103, 77.34392 | 30 KM/Hr    |
| 46 | Turn       | High       | 10.59171, 77.20777 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 6  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 10 | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |

|    | type     | name                          | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------|------------------------|-------------|------------|
| 11 | hospital | Government Hospital           | 11.2343326, 77.719226  | 30 km/h     | Medium     |
| 14 | police   | Kangeyam Police Station       | 11.0011411, 77.562688  | 30 km/h     | Medium     |
| 15 | hospital | J K Hospital                  | 11.0042298, 77.5603345 | 30 km/h     | Medium     |
| 18 | hospital | Sri Balaji Surya Hospital     | 10.735518, 77.5185388  | 30 km/h     | Medium     |
| 19 | hospital | Ramana Hospital, Dharapuram   | 10.7351092, 77.517107  | 30 km/h     | Medium     |
| 22 | hospital | Janani Hospital, Dharapuram   | 10.7294033, 77.5210119 | 30 km/h     | Medium     |
| 23 | clinic   | Punitha Nursing Home          | 10.72901, 77.520148    | 30 km/h     | Medium     |
| 24 | hospital | Ponnaiah Hospital, Dharapuram | 10.72916, 77.520131    | 30 km/h     | Medium     |
| 25 | pharmacy | Apollo Pharmacy               | 10.5860212, 77.2456546 | 30 km/h     | Medium     |
| 26 | hospital | Anitha Hospital               | 10.58568, 77.248727    | 30 km/h     | Medium     |
| 27 | hospital | ARVI Hospital                 | 10.58528, 77.249095    | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                                 | coordinates            | speed_limit | risk_level |
|----|-------------|------------------------------------------------------|------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                         | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                        | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை                  | 11.3833608, 77.8970145 | 30 km/h     | Medium     |
| 9  | school      | KSR Educational institution                          | 11.3772013, 77.8908807 | 30 km/h     | Medium     |
| 12 | college     | Government Arts and Science College, Kangeyam        | 11.0626738, 77.636227  | 30 km/h     | Medium     |
| 13 | school      | Carmel Girls Higher Secondary School                 | 11.0009019, 77.5710182 | 30 km/h     | Medium     |
| 16 | school      | Global Matriculation Higher Secondary School         | 10.965994, 77.5591898  | 30 km/h     | Medium     |
| 17 | college     | Maharani College                                     | 10.7625655, 77.5157126 | 30 km/h     | Medium     |
| 20 | school      | Vishnu Lakshmi Matriculation Higher Secondary School | 10.7340392, 77.5164506 | 30 km/h     | Medium     |



|    | type   | name                              | coordinates               | speed_limit | risk_level |
|----|--------|-----------------------------------|---------------------------|-------------|------------|
| 21 | school | Thenmalar Higher Secondary School | 10.7333748,<br>77.5166481 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.00172, 77.56268





**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.72668, 77.52217



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.58763, 77.24722





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44022, 77.87546



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.43107, 77.88171



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.39976, 77.88382





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.39690, 77.88852



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.36593, 77.88847





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.33561, 77.86176



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.31790, 77.84330



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.30951, 77.77598





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.31109, 77.77183



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.30911, 77.76654





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29496, 77.74717



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29128, 77.74656



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28859, 77.74323





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.28533, 77.74247



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.27310, 77.72938





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.27330, 77.72720



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.23062, 77.71799



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.15942, 77.70068



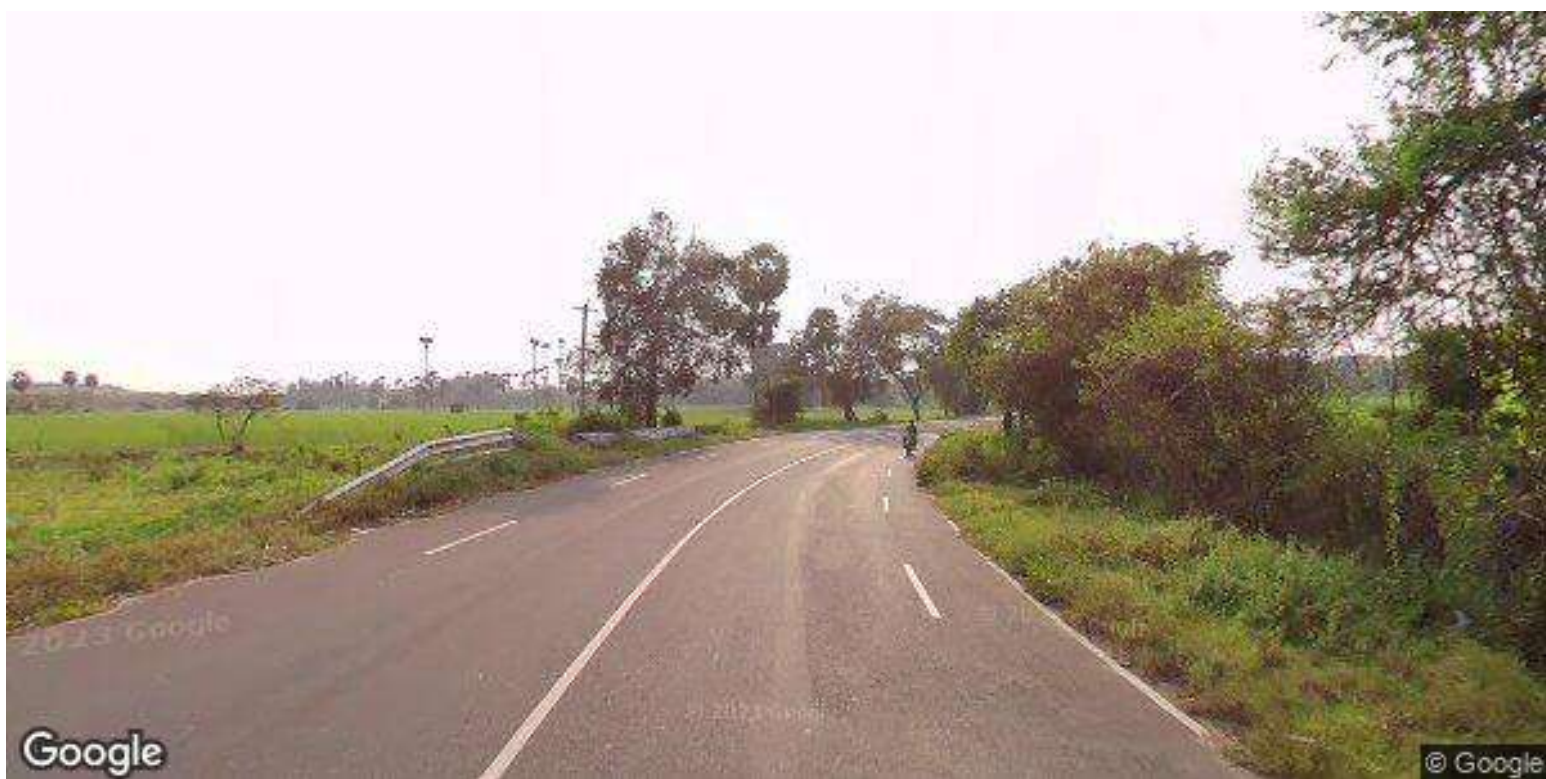


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.12126, 77.67929



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.11935, 77.67790





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.11765, 77.67237



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.11660, 77.67164



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.10462, 77.67055





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08701, 77.66412



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.06229, 77.63986





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.00834, 77.56962



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.00065, 77.56884



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.94873, 77.54679





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.94732, 77.54710



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.93774, 77.53723





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.92953, 77.53274



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.89225, 77.52651



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.83362, 77.52899





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.77658, 77.50715



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.75748, 77.51939





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.73682, 77.51732



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.63697, 77.41792



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.63715, 77.40497





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.62891, 77.38715



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.62505, 77.37856





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.63103, 77.34392



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.59171, 77.20777

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